SIS SAMPLE DRAFT — FULL TIER

Albany County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 18, 2026 · 42.7100, -73.8200

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Albany County Sample — Shopping Center / Retail
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within Albany-Schenectady-Troy MSA.
NYSDOT Region	Region 1 — Capital District
Site coordinates	42.7100°, -73.8200°
Opening year (ETC)	2027

Note:

It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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PROJECT CONTEXT

NYSDOT region: Region 1 — Capital District
Planning group: CDTC (Capital District Transportation Committee)
Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).
SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required; Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required; Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required; Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

GML §239-m / §239-n county planning board referral required:

The site is within 500 ft of Wolf Road (Urban Principal Arterial Other), which triggers mandatory referral to the controlling county planning board under General Municipal Law §239-m (site-plan / special-permit / variance / zoning amendment) or §239-n (subdivision review) before the local board may act. The county may make adverse findings under §239-f; the local board can override only by supermajority. Practical TIS consequence: the county will typically request the TIS in full as part of the §239 referral package — pre-application coordination with the county planning agency is recommended.

AT LOS E/F

12.6s

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (https://gis.dot.ny.gov/html5viewer/?viewer=tdv). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Near Wolf Road (Wolf Road)	Urban Principal Arterial Other	40 mph	Speed study required
Wolf Road (Wolf Road)	Urban Principal Arterial Other	40 mph	Speed study required
Near Central Avenue	Urban Principal Arterial Other	40 mph	Speed study required
Near Wolf Road (Wolf Road)	Urban Principal Arterial Other	40 mph	Speed study required
Fuller Road	Urban Principal Arterial Other	40 mph	Speed study required
Near Central Avenue	Urban Principal Arterial Other	40 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Principal Arterial Other (8); Urban Principal Arterial Interstate (1); Urban Local (1); Urban Minor Arterial (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component v/c > 1.0 is LOS F regardless of delay or density.

Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any v/c > 1.0 is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10–20	20–35	35–55	55–80	>80

Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any v/c > 1.0 is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10–15	15–25	25–35	35–50	>50

Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component vd/c > 1.00 is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤11	11–18	18–26	26–35	35–45	>45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 1 — Capital District: 0.02%/yr median across 2,050 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.42% to 1.28%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional

Planning Group CDTC (Capital District Transportation Committee) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Near Wolf Road	12,780	1,278	12,972	1,297	17,471	1,747	20,276	2,028
Wolf Road & Central Avenue	12,779	1,278	12,971	1,297	17,470	1,747	20,274	2,027
Near Central Avenue	14,620	1,462	14,839	1,484	19,986	1,999	23,195	2,320
Near Wolf Road	12,780	1,278	12,972	1,297	17,471	1,747	20,276	2,028
Fuller Road & Central Avenue	14,620	1,462	14,839	1,484	19,986	1,999	23,195	2,320
Near Central Avenue	5,369	537	5,450	545	7,340	734	8,518	852
Wolf Road & Sand Creek Road	11,171	1,117	11,339	1,134	15,271	1,527	17,723	1,772
Sand Creek Road & Adirondack Northway	6,210	621	6,303	630	8,489	849	9,852	985
Near Sand Creek Road	4,530	453	4,598	460	6,193	619	7,187	719
Lincoln Avenue & Central Avenue	39,150	3,915	39,737	3,974	53,520	5,352	62,113	6,211
Near Sand Creek Road	6,209	621	6,302	630	8,488	849	9,851	985

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
Near Wolf Road	Signal (controlling RTO to confirm)	Field verification required
Wolf Road & Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Near Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Near Wolf Road	Signal (controlling RTO to confirm)	Field verification required
Fuller Road & Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Near Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Wolf Road & Sand Creek Road	Signal (controlling RTO to confirm)	Field verification required
Sand Creek Road & Adirondack Northway	Signal (controlling RTO to confirm)	Field verification required
Near Sand Creek Road	Signal (controlling RTO to confirm)	Field verification required
Lincoln Avenue & Central Avenue	Signal (controlling RTO to confirm)	Field verification required
Near Sand Creek Road	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Near Wolf Road	C	25.2	C	25.6	0.72
Wolf Road & Central Avenue	C	25.2	C	25.6	0.72
Near Central Avenue	C	30.2	C	31.0	0.82
Near Wolf Road	C	25.2	C	25.6	0.72
Fuller Road & Central Avenue	C	30.2	C	31.0	0.82
Near Central Avenue	B	16.7	B	16.7	0.30
Wolf Road & Sand Creek Road	C	22.4	C	22.7	0.63
Sand Creek Road & Adirondack Northway	B	17.3	B	17.4	0.35
Near Sand Creek Road	B	16.1	B	16.1	0.26
Lincoln Avenue & Central Avenue	F	120.0	F	120.0	2.21
Near Sand Creek Road	B	17.3	B	17.4	0.35

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

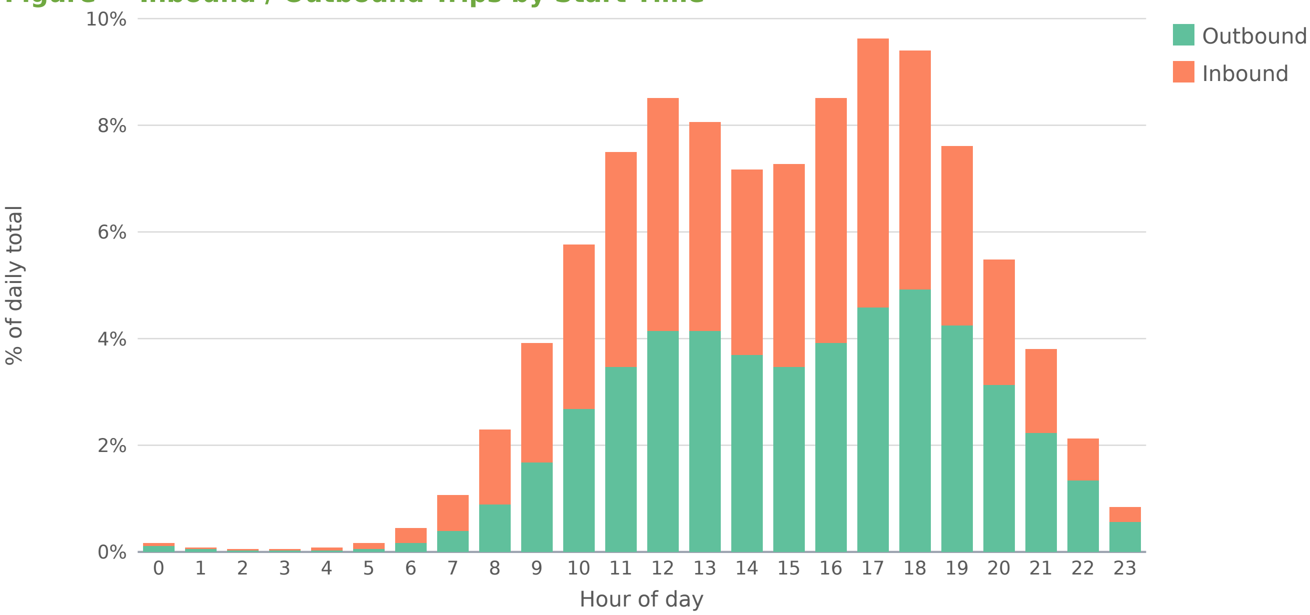
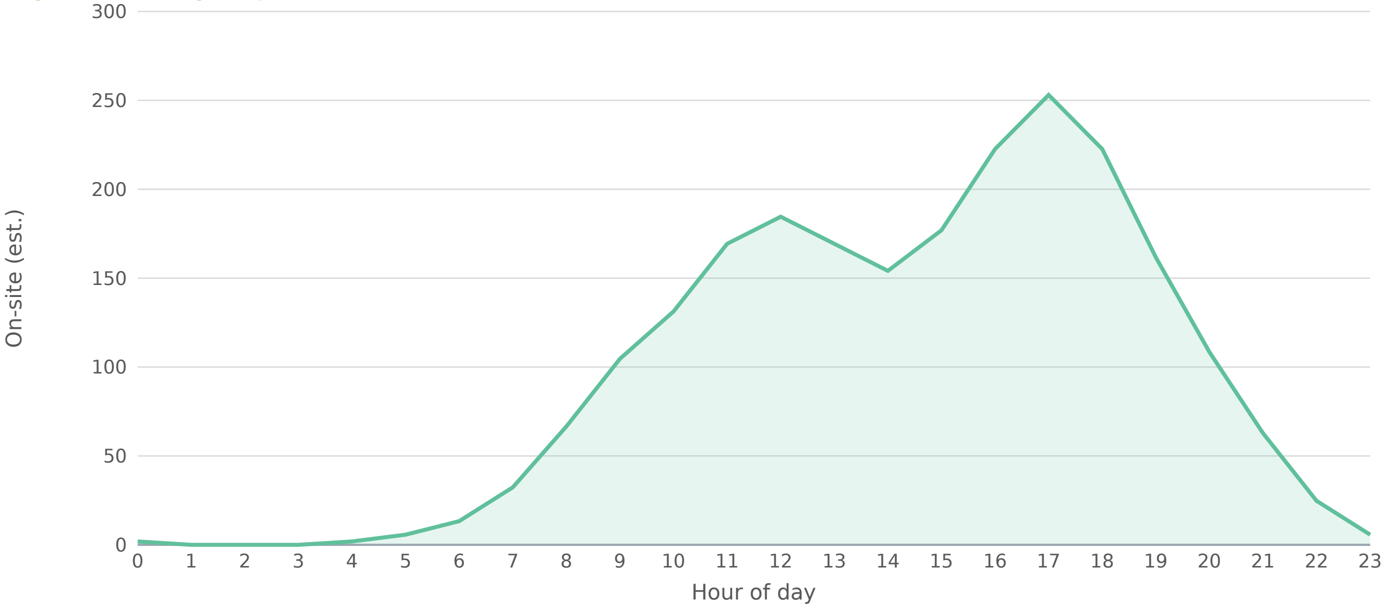


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Near Wolf Road	C	▲ D	D	F	12.6
Wolf Road & Central Avenue	C	C	D	F	6.4
Near Central Avenue	C	▲ D	F	F	8.4
Near Wolf Road	C	C	D	E	2.9
Fuller Road & Central Avenue	C	C	F	F	1.5
Near Central Avenue	B	B	B	B	0.2
Wolf Road & Sand Creek Road	C	C	C	C	0.4
Sand Creek Road & Adirondack Northway	B	B	B	B	0.1
Near Sand Creek Road	B	B	B	B	0.2
Lincoln Avenue & Central Avenue	F	F	F	F	0.0
Near Sand Creek Road	B	B	B	B	0.1

3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

Near Wolf Road [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Wolf Road & Central Avenue [MINOR]

Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Near Central Avenue [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Lincoln Avenue & Central Avenue [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

3.7 Trip Distribution — Gravity Model

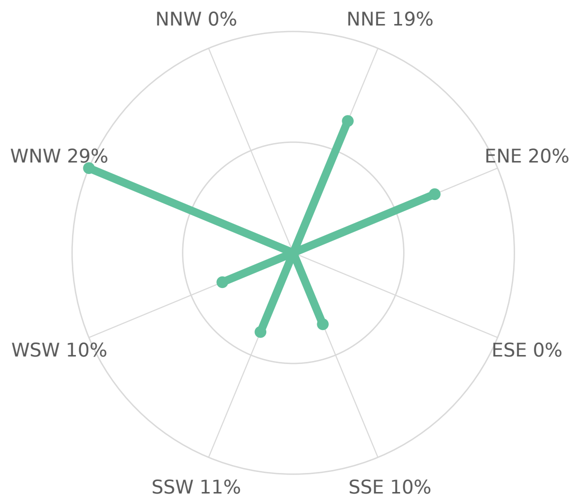
The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	39%
Southeast (ESE + SSE)	10%
Southwest (SSW + WSW)	21%
Northwest (WNW + NNW)	29%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Lincoln Avenue & Central Avenue	WNW	0.66	3,915	0.3	25.53%
Near Central Avenue	SSW	0.21	1,462	0.1	11.32%
Near Wolf Road	NNE	0.06	1,278	0.1	10.59%
Fuller Road & Central Avenue	SSE	0.50	1,462	0.1	10.21%
Wolf Road & Central Avenue	WSW	0.15	1,278	0.1	10.14%
Near Wolf Road	ENE	0.27	1,278	0.1	9.66%
Wolf Road & Sand Creek Road	ENE	0.59	1,117	0.1	7.57%
Sand Creek Road & Adirondack Northway	NNE	0.60	621	0.0	4.19%
Near Sand Creek Road	NNE	0.69	621	0.0	4.07%
Near Central Avenue	WNW	0.54	537	0.0	3.70%
Near Sand Creek Road	ENE	0.65	453	0.0	3.01%

11 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

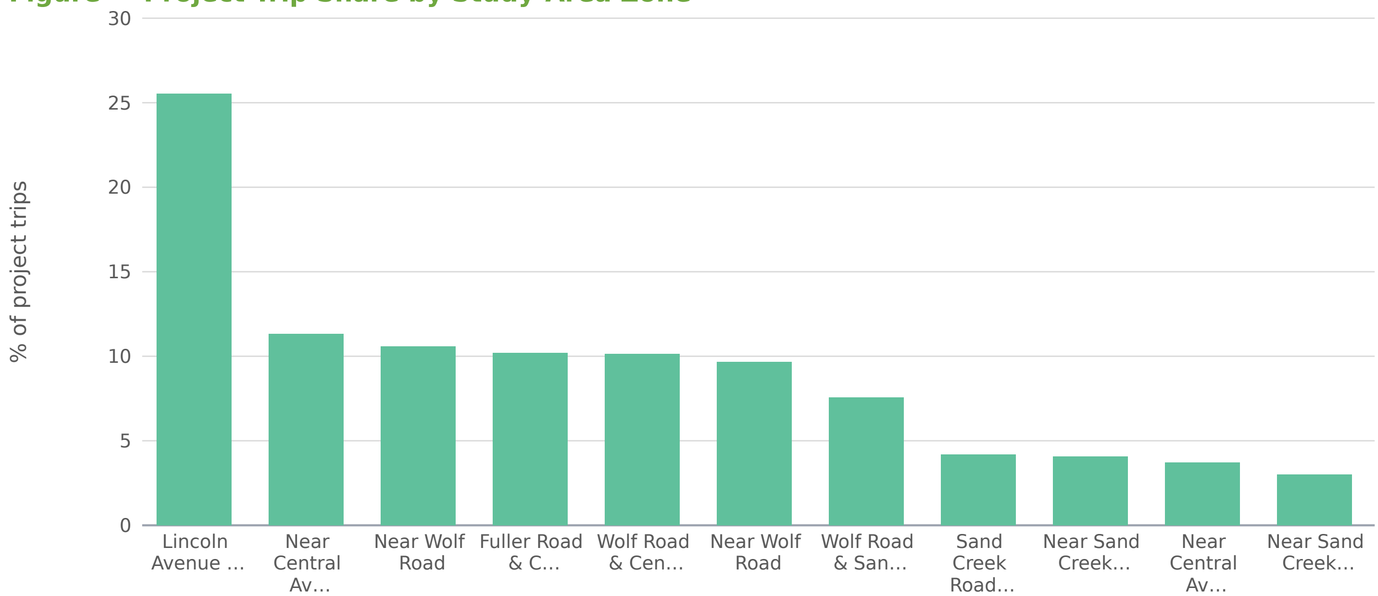
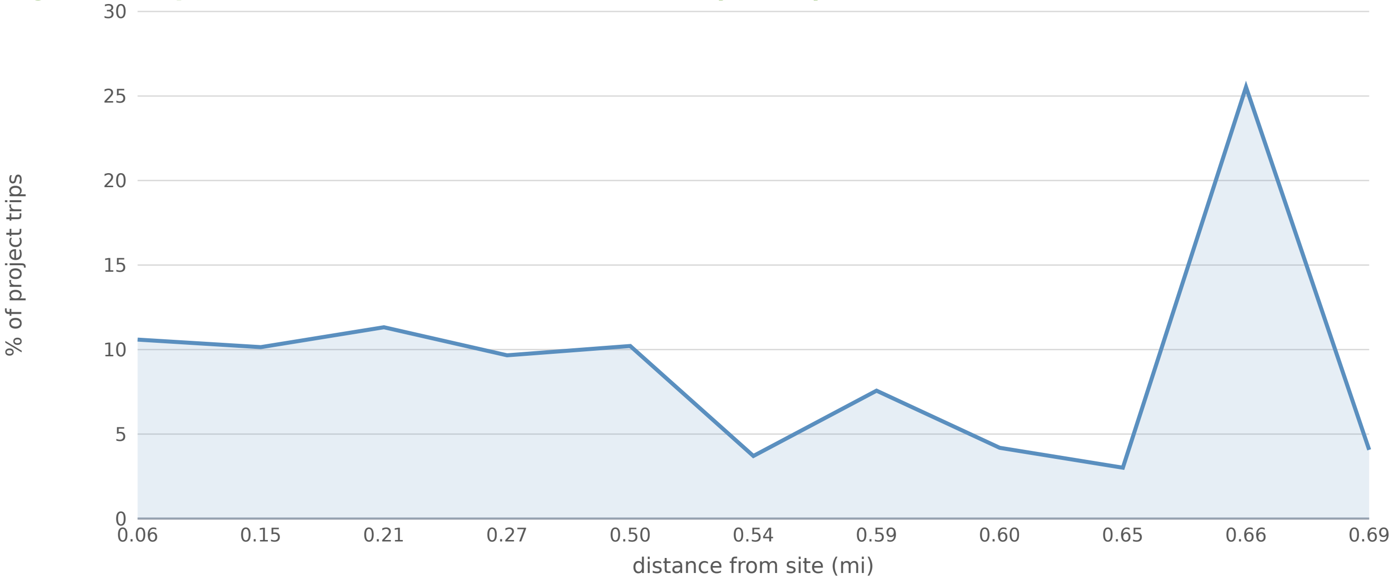


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near Wolf Road	NNE	0.06	173	328	33.7%
Wolf Road & Central Avenue	WSW	0.15	112	213	21.9%
Near Central Avenue	SSW	0.21	83	157	16.1%
Near Wolf Road	ENE	0.27	61	115	11.8%
Fuller Road & Central Avenue	SSE	0.50	20	37	3.8%
Near Central Avenue	WNW	0.54	16	31	3.2%
Wolf Road & Sand Creek Road	ENE	0.59	12	23	2.4%
Sand Creek Road & Adirondack Northway	NNE	0.60	11	22	2.3%
Near Sand Creek Road	ENE	0.65	9	17	1.7%
Lincoln Avenue & Central Avenue	WNW	0.66	9	17	1.7%
Near Sand Creek Road	NNE	0.69	7	14	1.4%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes – Case Information: Three Year Window" Open Data dataset (https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe), filtered by county_name resolved from the project site coordinates via the NYSDOT RDM_Roadway_Current FeatureServer. The values cover the entire rolling 3-year reporting window for ALBANY County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Fatal Accident	82	27.3
Injury Accident	1,028	342.7
Property Damage Accident	29,867	9955.7
Property Damage & Injury Accident	6,426	2142.0
Total (all severities)	37,403	12467.7

Within ALBANY County over the rolling 3-year window, 37,403 crashes were reported to NY DMV (82 fatal, 7,454 injury-involved, 29,867 property damage only). The Fatal share is 0.073% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 1 — Capital District Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits. Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Wolf Road	NB	413	420	590
Near Wolf Road	SB	315	319	398
Near Wolf Road	EB	325	330	330
Near Wolf Road	WB	225	229	308
Wolf Road & Central Avenue	NB	368	374	406
Wolf Road & Central Avenue	SB	344	349	349
Wolf Road & Central Avenue	EB	290	295	365
Wolf Road & Central Avenue	WB	276	280	390
Near Central Avenue	NB	422	428	487
Near Central Avenue	SB	366	372	453
Near Central Avenue	EB	339	345	361
Near Central Avenue	WB	335	340	340
Near Wolf Road	NB	371	376	376
Near Wolf Road	SB	336	341	367
Near Wolf Road	EB	341	346	406
Near Wolf Road	WB	231	234	263
Fuller Road & Central Avenue	NB	386	392	410
Fuller Road & Central Avenue	SB	403	409	428
Fuller Road & Central Avenue	EB	400	406	406
Fuller Road & Central Avenue	WB	273	278	278
Near Central Avenue	NB	147	149	149
Near Central Avenue	SB	120	122	123
Near Central Avenue	EB	154	157	170
Near Central Avenue	WB	116	118	134
Wolf Road & Sand Creek Road	NB	365	371	371
Wolf Road & Sand Creek Road	SB	291	295	300
Wolf Road & Sand Creek Road	EB	275	279	291
Wolf Road & Sand Creek Road	WB	187	190	195
Sand Creek Road & Adirondack Northway	NB	163	165	177
Sand Creek Road & Adirondack Northway	SB	155	157	163
Sand Creek Road & Adirondack Northway	EB	180	183	183
Sand Creek Road & Adirondack Northway	WB	123	125	130
Near Sand Creek Road	NB	152	155	155
Near Sand Creek Road	SB	113	114	117
Near Sand Creek Road	EB	105	107	116

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Sand Creek Road	WB	83	84	90
Lincoln Avenue & Central Avenue	NB	1,116	1,133	1,133
Lincoln Avenue & Central Avenue	SB	1,116	1,133	1,134
Lincoln Avenue & Central Avenue	EB	924	938	945
Lincoln Avenue & Central Avenue	WB	758	770	779
Near Sand Creek Road	NB	181	183	191
Near Sand Creek Road	SB	150	152	156
Near Sand Creek Road	EB	168	170	171
Near Sand Creek Road	WB	123	125	127

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Wolf Road	NB	420	0.52	20.1	C
Near Wolf Road	SB	319	0.39	18.0	B
Near Wolf Road	EB	330	0.41	18.2	B
Near Wolf Road	WB	229	0.28	16.5	B
Wolf Road & Central Avenue	NB	374	0.46	19.1	B
Wolf Road & Central Avenue	SB	349	0.43	18.6	B
Wolf Road & Central Avenue	EB	295	0.36	17.5	B
Wolf Road & Central Avenue	WB	280	0.35	17.3	B
Near Central Avenue	NB	428	0.53	20.3	C
Near Central Avenue	SB	372	0.46	19.0	B
Near Central Avenue	EB	345	0.43	18.5	B
Near Central Avenue	WB	340	0.42	18.4	B
Near Wolf Road	NB	376	0.46	19.1	B
Near Wolf Road	SB	341	0.42	18.4	B
Near Wolf Road	EB	346	0.43	18.5	B
Near Wolf Road	WB	234	0.29	16.5	B
Fuller Road & Central Avenue	NB	392	0.48	19.5	B
Fuller Road & Central Avenue	SB	409	0.50	19.9	B
Fuller Road & Central Avenue	EB	406	0.50	19.8	B
Fuller Road & Central Avenue	WB	278	0.34	17.2	B
Near Central Avenue	NB	149	0.18	15.3	B
Near Central Avenue	SB	122	0.15	15.0	B
Near Central Avenue	EB	157	0.19	15.4	B
Near Central Avenue	WB	118	0.15	14.9	B
Wolf Road & Sand Creek Road	NB	371	0.46	19.0	B
Wolf Road & Sand Creek Road	SB	295	0.36	17.5	B
Wolf Road & Sand Creek Road	EB	279	0.34	17.3	B
Wolf Road & Sand Creek Road	WB	190	0.23	15.9	B
Sand Creek Road & Adirondack Northway	NB	165	0.20	15.6	B
Sand Creek Road & Adirondack Northway	SB	157	0.19	15.4	B
Sand Creek Road & Adirondack Northway	EB	183	0.23	15.8	B
Sand Creek Road & Adirondack Northway	WB	125	0.15	15.0	B
Near Sand Creek Road	NB	155	0.19	15.4	B
Near Sand Creek Road	SB	114	0.14	14.9	B
Near Sand Creek Road	EB	107	0.13	14.8	B
Near Sand Creek Road	WB	84	0.10	14.5	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Lincoln Avenue & Central Avenue	NB	1,133	1.40	120.0	F
Lincoln Avenue & Central Avenue	SB	1,133	1.40	120.0	F
Lincoln Avenue & Central Avenue	EB	938	1.16	109.5	F
Lincoln Avenue & Central Avenue	WB	770	0.95	45.4	D
Near Sand Creek Road	NB	183	0.23	15.8	B
Near Sand Creek Road	SB	152	0.19	15.4	B
Near Sand Creek Road	EB	170	0.21	15.6	B
Near Sand Creek Road	WB	125	0.15	15.0	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Wolf Road	NB	590	0.73	25.9	C
Near Wolf Road	SB	398	0.49	19.6	B
Near Wolf Road	EB	330	0.41	18.2	B
Near Wolf Road	WB	308	0.38	17.8	B
Wolf Road & Central Avenue	NB	406	0.50	19.8	B
Wolf Road & Central Avenue	SB	349	0.43	18.6	B
Wolf Road & Central Avenue	EB	365	0.45	18.9	B
Wolf Road & Central Avenue	WB	390	0.48	19.4	B
Near Central Avenue	NB	487	0.60	22.0	C
Near Central Avenue	SB	453	0.56	21.0	C
Near Central Avenue	EB	361	0.45	18.8	B
Near Central Avenue	WB	340	0.42	18.4	B
Near Wolf Road	NB	376	0.46	19.1	B
Near Wolf Road	SB	367	0.45	18.9	B
Near Wolf Road	EB	406	0.50	19.8	B
Near Wolf Road	WB	263	0.32	17.0	B
Fuller Road & Central Avenue	NB	410	0.51	19.9	B
Fuller Road & Central Avenue	SB	428	0.53	20.3	C
Fuller Road & Central Avenue	EB	406	0.50	19.8	B
Fuller Road & Central Avenue	WB	278	0.34	17.2	B
Near Central Avenue	NB	149	0.18	15.3	B
Near Central Avenue	SB	123	0.15	15.0	B
Near Central Avenue	EB	170	0.21	15.6	B
Near Central Avenue	WB	134	0.17	15.1	B
Wolf Road & Sand Creek Road	NB	371	0.46	19.0	B
Wolf Road & Sand Creek Road	SB	300	0.37	17.6	B
Wolf Road & Sand Creek Road	EB	291	0.36	17.5	B
Wolf Road & Sand Creek Road	WB	195	0.24	16.0	B
Sand Creek Road & Adirondack Northway	NB	177	0.22	15.7	B
Sand Creek Road & Adirondack Northway	SB	163	0.20	15.5	B
Sand Creek Road & Adirondack Northway	EB	183	0.23	15.8	B
Sand Creek Road & Adirondack Northway	WB	130	0.16	15.1	B
Near Sand Creek Road	NB	155	0.19	15.4	B
Near Sand Creek Road	SB	117	0.14	14.9	B
Near Sand Creek Road	EB	116	0.14	14.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Sand Creek Road	WB	90	0.11	14.6	B
Lincoln Avenue & Central Avenue	NB	1,133	1.40	120.0	F
Lincoln Avenue & Central Avenue	SB	1,134	1.40	120.0	F
Lincoln Avenue & Central Avenue	EB	945	1.17	113.0	F
Lincoln Avenue & Central Avenue	WB	779	0.96	47.4	D
Near Sand Creek Road	NB	191	0.24	15.9	B
Near Sand Creek Road	SB	156	0.19	15.4	B
Near Sand Creek Road	EB	171	0.21	15.6	B
Near Sand Creek Road	WB	127	0.16	15.1	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: a=28507, b=-0.02, c=-0.123) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Wolf Road	0.06	0.1	328	33.7%
Wolf Road & Central Avenue	0.15	0.4	213	21.9%
Near Central Avenue	0.21	0.5	157	16.1%
Near Wolf Road	0.27	0.6	115	11.8%
Fuller Road & Central Avenue	0.50	1.2	37	3.8%
Near Central Avenue	0.54	1.3	31	3.2%
Wolf Road & Sand Creek Road	0.59	1.4	23	2.4%
Sand Creek Road & Adirondack Northway	0.60	1.4	22	2.3%
Near Sand Creek Road	0.65	1.6	17	1.7%
Lincoln Avenue & Central Avenue	0.66	1.6	17	1.7%
Near Sand Creek Road	0.69	1.7	14	1.4%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 2.22. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

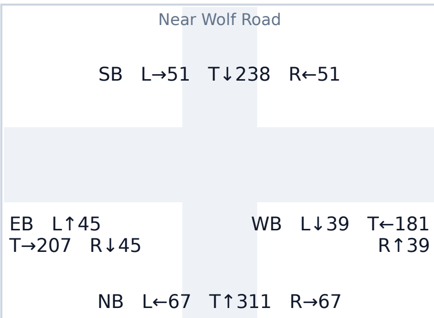
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near Wolf Road

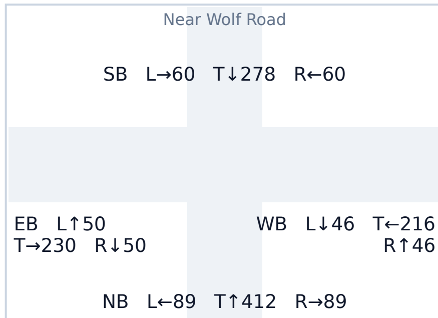
Signal ID	albany-951
Location	Central Albany-Schenectady-Troy · 0.06 mi from site
Added PM peak trips	328
Existing / No-Build (PM)	LOS C · 25.6 s/veh · v/c 0.72
Build (PM)	LOS D · 38.2 s/veh · v/c 0.90
Δ control delay	+12.6 s/veh (LOS change)
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

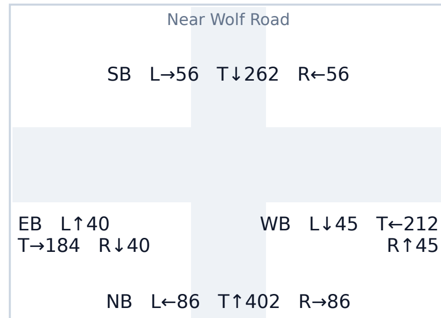
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	420	171	590	0.52 → 0.73	20.1 → 25.9	C → C	498
SB	319	78	398	0.39 → 0.49	18.0 → 19.6	B → B	289
EB	330	0	330	0.41 → 0.41	18.2 → 18.2	B → B	229
WB	229	79	308	0.28 → 0.38	16.5 → 17.8	B → B	210

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Right	86	26%
NB Through	85	26%
WB Left	79	24%
SB Through	78	24%

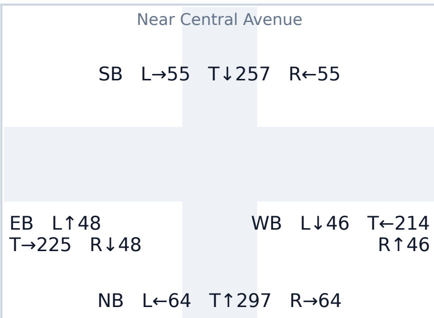
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Near Central Avenue

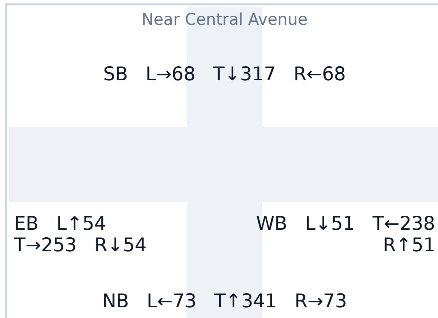
Signal ID	albany-33
Location	Central Albany-Schenectady-Troy · 0.21 mi from site
Added PM peak trips	157
Existing / No-Build (PM)	LOS C · 31.0 s/veh · v/c 0.82
Build (PM)	LOS D · 39.4 s/veh · v/c 0.91
Δ control delay	+8.4 s/veh (LOS change)
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

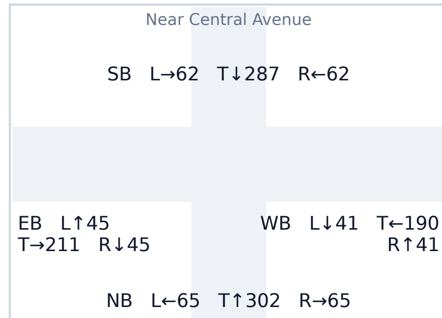
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	428	59	487	0.53 → 0.60	20.3 → 22.0	C → C	379
SB	372	82	453	0.46 → 0.56	19.0 → 21.0	B → C	344
EB	345	16	361	0.43 → 0.45	18.5 → 18.8	B → B	256
WB	340	0	340	0.42 → 0.42	18.4 → 18.4	B → B	238

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	64	41%
NB Through	59	38%
SB Right	18	11%
EB Left	16	10%

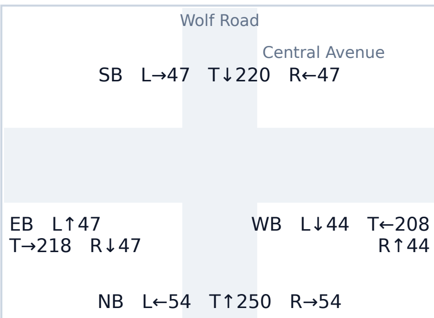
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Wolf Road & Central Avenue

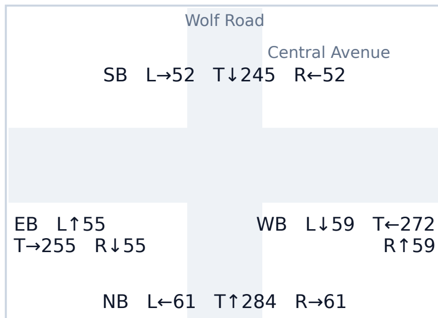
Signal ID	albany-1163
Location	Central Albany-Schenectady-Troy · 0.15 mi from site
Added PM peak trips	213
Existing / No-Build (PM)	LOS C · 25.6 s/veh · v/c 0.72
Build (PM)	LOS C · 32.0 s/veh · v/c 0.84
Δ control delay	+6.4 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

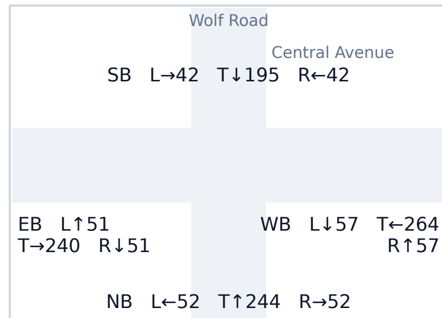
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	374	32	406	0.46 → 0.50	19.1 → 19.8	B → B	297
SB	349	0	349	0.43 → 0.43	18.6 → 18.6	B → B	246
EB	295	70	365	0.36 → 0.45	17.5 → 18.9	B → B	259
WB	280	111	390	0.35 → 0.48	17.3 → 19.4	B → B	283

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	76	36%
EB Through	70	33%
WB Left	35	16%
NB Right	32	15%

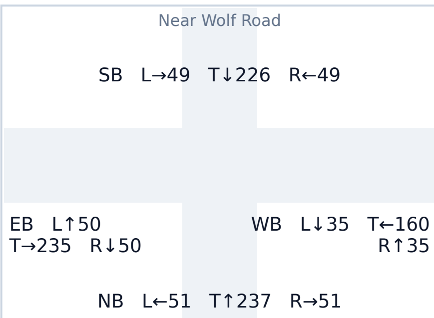
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Near Wolf Road

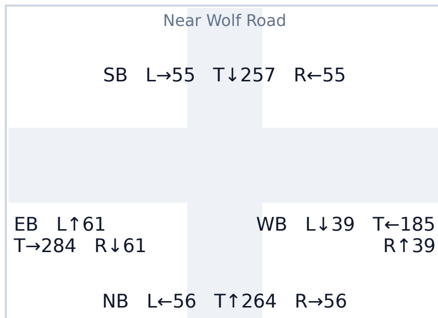
Signal ID	albany-950
Location	Central Albany-Schenectady-Troy · 0.27 mi from site
Added PM peak trips	115
Existing / No-Build (PM)	LOS C · 25.6 s/veh · v/c 0.72
Build (PM)	LOS C · 28.5 s/veh · v/c 0.78
Δ control delay	+2.9 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

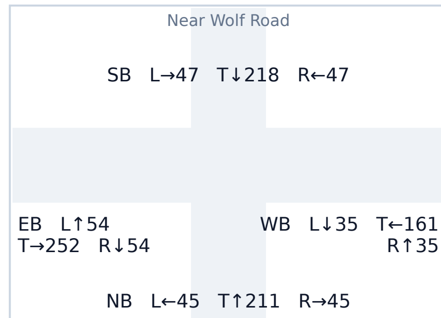
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	376	0	376	0.46 → 0.46	19.1 → 19.1	B → B	270
SB	341	26	367	0.42 → 0.45	18.4 → 18.9	B → B	261
EB	346	60	406	0.43 → 0.50	18.5 → 19.8	B → B	297
WB	234	29	263	0.29 → 0.32	16.5 → 17.0	B → B	175

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	32	28%
WB Through	29	25%
EB Left	28	24%
SB Right	26	23%

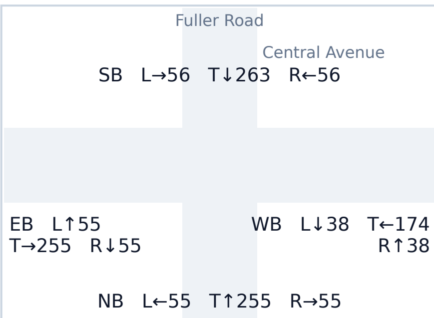
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Fuller Road & Central Avenue

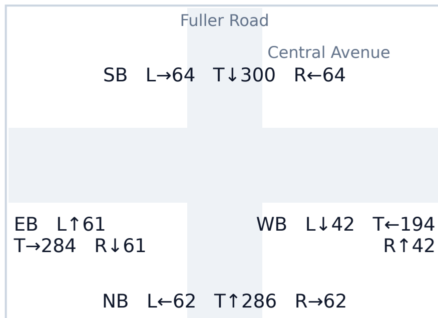
Signal ID	albany-415
Location	Central Albany-Schenectady-Troy · 0.50 mi from site
Added PM peak trips	37
Existing / No-Build (PM)	LOS C · 31.0 s/veh · v/c 0.82
Build (PM)	LOS C · 32.5 s/veh · v/c 0.84
Δ control delay	+1.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

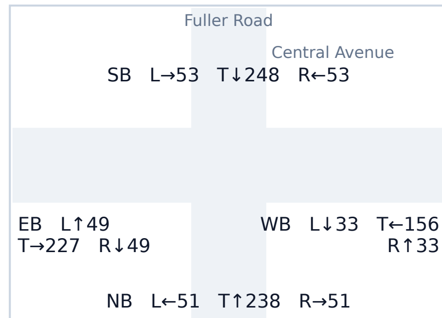
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	392	18	410	0.48 → 0.51	19.5 → 19.9	B → B	301
SB	409	19	428	0.50 → 0.53	19.9 → 20.3	B → C	318
EB	406	0	406	0.50 → 0.50	19.8 → 19.8	B → B	297
WB	278	0	278	0.34 → 0.34	17.2 → 17.2	B → B	186

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	19	51%
NB Through	18	49%

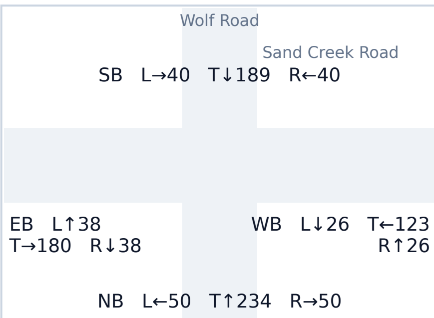
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Wolf Road & Sand Creek Road

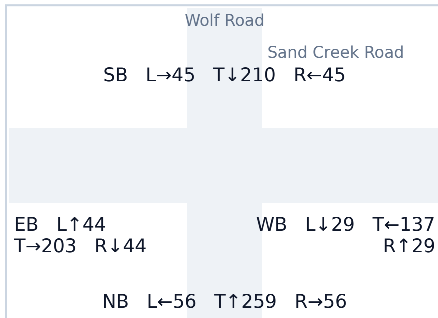
Signal ID	albany-962
Location	Central Albany-Schenectady-Troy · 0.59 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS C · 22.7 s/veh · v/c 0.63
Build (PM)	LOS C · 23.1 s/veh · v/c 0.64
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

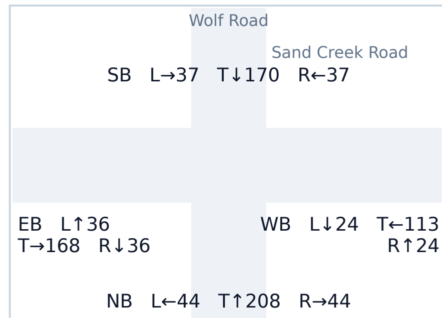
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	371	0	371	0.46 → 0.46	19.0 → 19.0	B → B	265
SB	295	5	300	0.36 → 0.37	17.5 → 17.6	B → B	204
EB	279	12	291	0.34 → 0.36	17.3 → 17.5	B → B	197
WB	190	6	195	0.23 → 0.24	15.9 → 16.0	B → B	124

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	6	26%
WB Through	6	26%
EB Left	6	26%
SB Right	5	22%

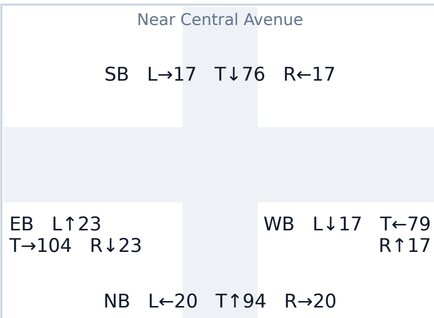
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Near Central Avenue

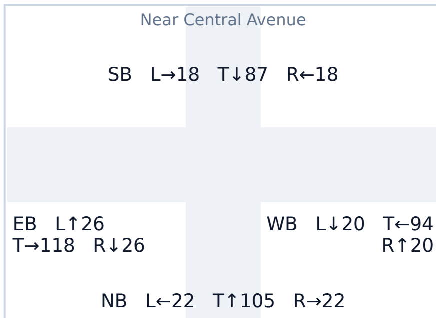
Signal ID	albany-119
Location	Central Albany-Schenectady-Troy · 0.54 mi from site
Added PM peak trips	31
Existing / No-Build (PM)	LOS B · 16.7 s/veh · v/c 0.30
Build (PM)	LOS B · 16.9 s/veh · v/c 0.32
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

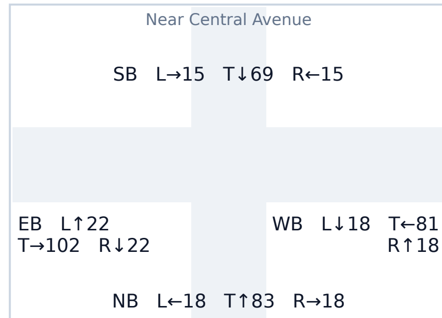
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	149	0	149	0.18 → 0.18	15.3 → 15.3	B → B	92
SB	122	1	123	0.15 → 0.15	15.0 → 15.0	B → B	75
EB	157	14	170	0.19 → 0.21	15.4 → 15.6	B → B	106
WB	118	16	134	0.15 → 0.17	14.9 → 15.1	B → B	82

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	15	48%
EB Through	14	45%
WB Right	1	3%
SB Left	1	3%

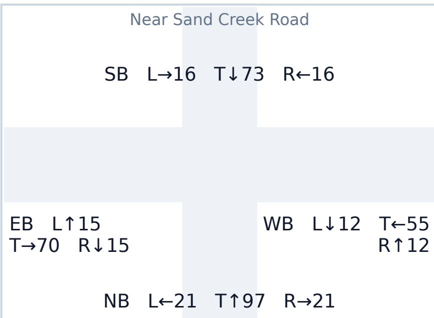
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Near Sand Creek Road

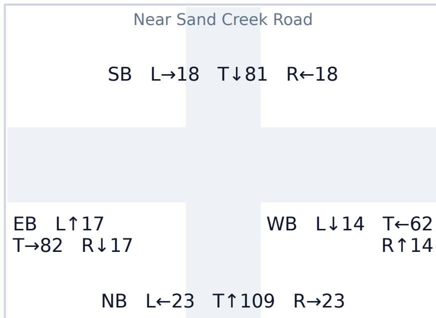
Signal ID	albany-14
Location	Central Albany-Schenectady-Troy · 0.65 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.26
Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

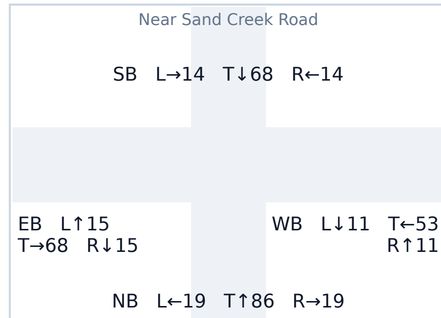
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	155	0	155	0.19 → 0.19	15.4 → 15.4	B → B	96
SB	114	3	117	0.14 → 0.14	14.9 → 14.9	B → B	71
EB	107	9	116	0.13 → 0.14	14.8 → 14.9	B → B	70
WB	84	5	90	0.10 → 0.11	14.5 → 14.6	B → B	53

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	6	35%
WB Through	5	29%
EB Left	3	18%
SB Right	3	18%

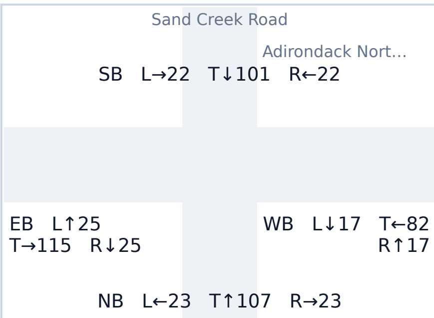
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Sand Creek Road & Adirondack Northway

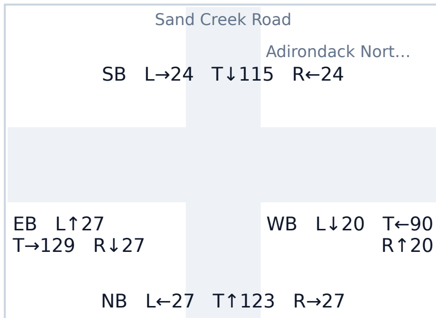
Signal ID	albany-961
Location	Central Albany-Schenectady-Troy · 0.60 mi from site
Added PM peak trips	22
Existing / No-Build (PM)	LOS B · 17.4 s/veh · v/c 0.35
Build (PM)	LOS B · 17.5 s/veh · v/c 0.36
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

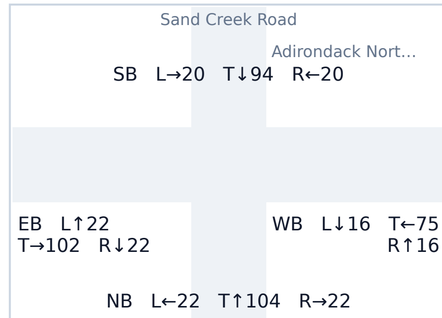
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	165	11	177	0.20 → 0.22	15.6 → 15.7	B → B	111
SB	157	6	163	0.19 → 0.20	15.4 → 15.5	B → B	102
EB	183	0	183	0.23 → 0.23	15.8 → 15.8	B → B	116
WB	125	5	130	0.15 → 0.16	15.0 → 15.1	B → B	79

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	6	27%
SB Through	6	27%
NB Right	5	23%
WB Left	5	23%

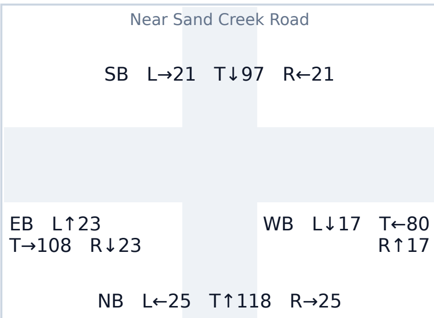
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Near Sand Creek Road

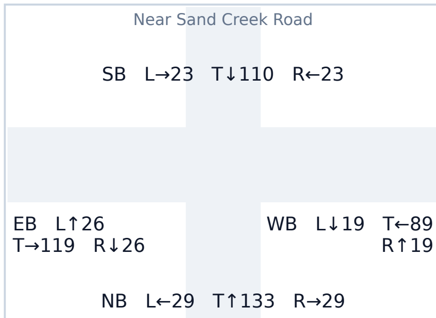
Signal ID	albany-451
Location	Central Albany-Schenectady-Troy · 0.69 mi from site
Added PM peak trips	14
Existing / No-Build (PM)	LOS B · 17.4 s/veh · v/c 0.35
Build (PM)	LOS B · 17.5 s/veh · v/c 0.36
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

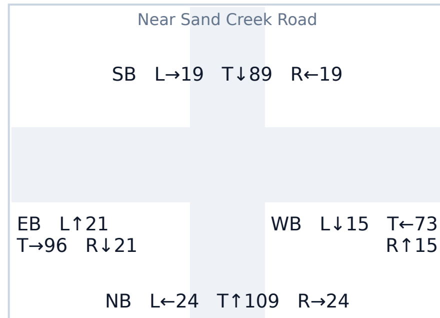
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	183	7	191	0.23 → 0.24	15.8 → 15.9	B → B	121
SB	152	4	156	0.19 → 0.19	15.4 → 15.4	B → B	97
EB	170	1	171	0.21 → 0.21	15.6 → 15.6	B → B	107
WB	125	2	127	0.15 → 0.16	15.0 → 15.1	B → B	77

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	4	29%
SB Through	4	29%
NB Right	2	14%
WB Left	2	14%
NB Left	1	7%
EB Right	1	7%

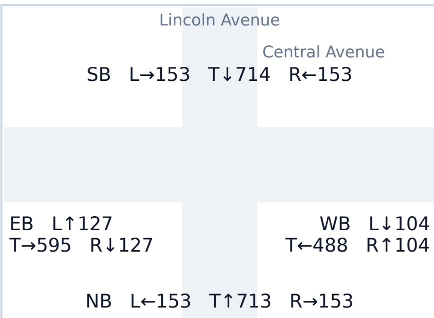
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Lincoln Avenue & Central Avenue

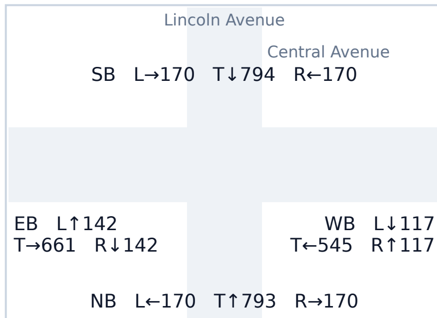
Signal ID	albany-118
Location	Central Albany-Schenectady-Troy · 0.66 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 2.21
Build (PM)	LOS F · 120.0 s/veh · v/c 2.22
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

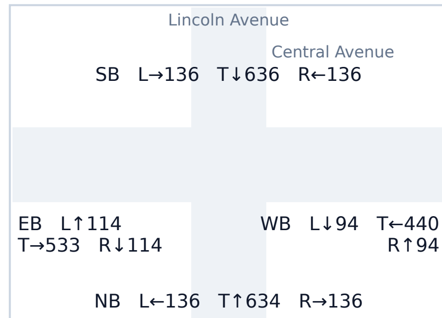
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,133	0	1,133	1.40 → 1.40	120.0 → 120.0	F → F	1,158
SB	1,133	1	1,134	1.40 → 1.40	120.0 → 120.0	F → F	1,160
EB	938	7	945	1.16 → 1.17	109.5 → 113.0	F → F	967
WB	770	9	779	0.95 → 0.96	45.4 → 47.4	D → D	778

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	8	47%
EB Through	7	41%
WB Right	1	6%
SB Left	1	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.